

Schroedinger's Graphics

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Synopsis

- Main Concept: Magic room with changing outside landscape
- Starting in a small room with a large window
- The landscape outside the window switches several time with button
- Each landscape has its own distinct characteristics and features
- Inspired by “Howl's Moving Castle”, particularly the magic dial that changes the scenery outside
- Main Goal: Implement various features that make each landscape rich and fun



image from “Howl's Moving Castle”
<https://www.youtube.com/watch?v=hHNGB9Wr2Y>

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Exemplar Image

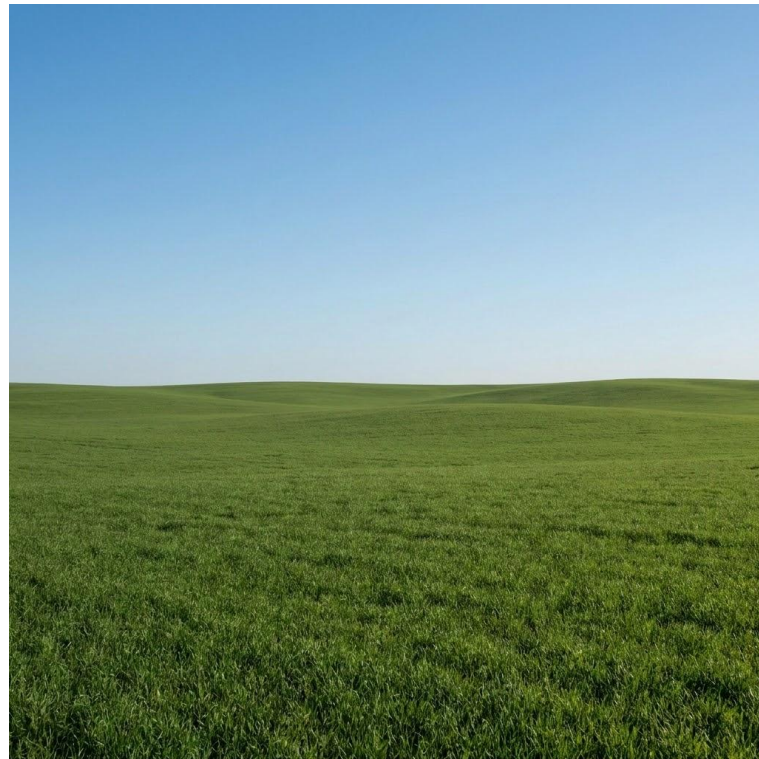


image generated by nano-banana
for scene explanation

Exemplar Image



Switch!



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List of Features

1. Shape-Guided Fire Particle Simulation
2. Mirage Effect (Desert Scene)
3. Skeletal Animation (Moving Fish)

Feature 1: Shape-Guided Fire Particle Simulation

- Goal
 - Develop a particle system where fire dynamically conforms to user-defined 3D geometries
- Key Technique
 - Curl Noise
 - Base fluid-like motion driven by Curl Noise
 - SDF-Gradient
 - Energy-minimizing guiding force smoothly pushes straying particles back or along the surface using the SDF gradient
- Reference
 - Particle: <https://learnopengl.com/In-Practice/2D-Game/Particles>
 - Collision Detection: <https://learnopengl.com/In-Practice/2D-Game/Collisions/Collision-detection>
 - Curl Noise: https://github.com/kbladin/Curl_Noise
 - SDF Collision: <https://docs.godotengine.org/en/stable/tutorials/3d/particles/collision.html>



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Feature 2: Mirage Effect

- Goal
 - Implement a rendering system where light paths dynamically curve to simulate authentic mirages.
- Key Technique
 - Hybrid Thermal Lattice Boltzmann Method
 - To express how the average number of flow particles due to interparticle interactions and ballistic motion
 - Non-linear ray casting
 - Solving the ray equation to trace curved light trajectories
- Reference
 - Mirage simulation using non-linear ray tracing:
<https://ieeexplore.ieee.org/document/4015408>
 - Particle: <https://learnopengl.com/In-Practice/2D-Game/Particles>



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Feature 3: Skeletal Animation

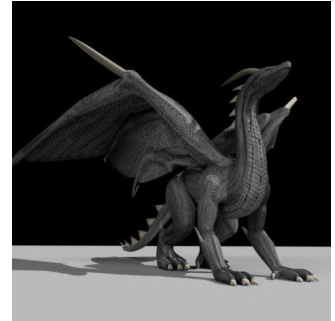
- Goal
 - Implement natural skeletal animation of moving fish with collada file
- Key Technique
 - Skeletal Animation
 - Manipulating hierarchical bone structure to express fish movement
 - SLERP for natural switching from scene to scene
 - Splines
 - For specifying moving path
- Reference
 - skeletal animation:
<https://learnopengl.com/Guest-Articles/2020/Skeletal-Animation>



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The Platform and the Assets

- Platform: OpenGL
- Assets:
 - Room: [free3d - Room](#)
 - Cat: [free3d - Cat](#)
 - Fish(.dae): [turbosquid - fish](#)
 - Shark(.dae): [turbosquid - shark](#)
 - Dragon: [free3d - dragon](#)
 - and more...



Extra

If time is enough...

- Lava flooding simulation
 - code: <https://github.com/Bixio999/3d-fluid-simulation>
- Aurora scene
 - report: https://graphics.stanford.edu/courses/cs348b-competition/cs348b-14/first_report.pdf
- Scene transition fade effect